UCTransNet: Rethinking the Skip Connections in U-Net from a Channel-wise Perspective with Transformer

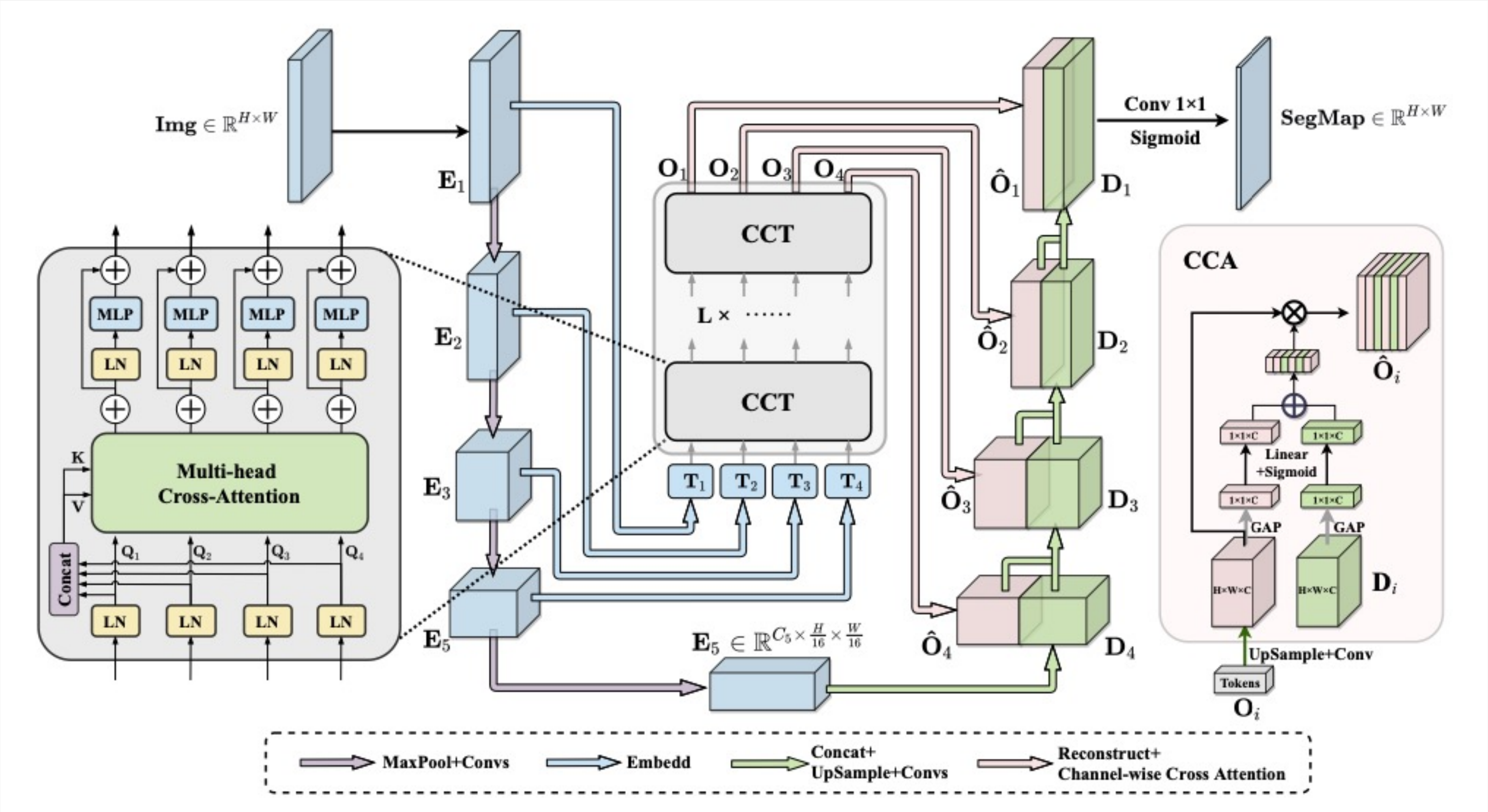
Haonan Wang, Peng Cao, Jiaqi Wang, Osmar R. Zaiane
Association for Advancement of Artificial Intelligence (AAAI), 2021

2022. 7. 2

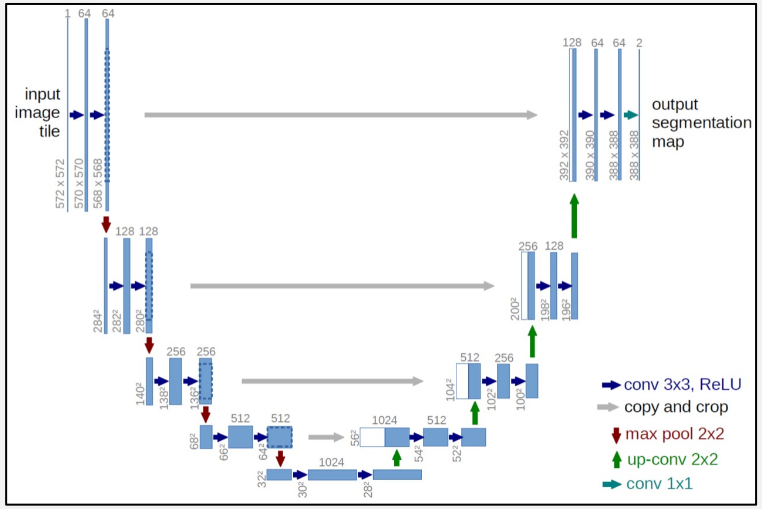
김준철, 김병현, 박재완, 이예은, 조경진, 이희재

Contents

Introduction Architecture



Introduction



- U-net

1. Encoder의 역할

 - Capture low-level and high-level features

2. Decoder 의 역할

 - Combine the semantic features to construct the final result

3. Skip connection 의 역할

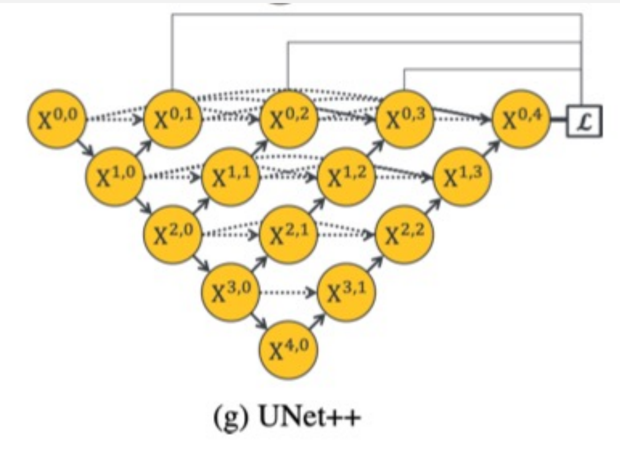
 - Help propagate spatial information to recover the full spatial resolution

- There exist Two semantic gaps

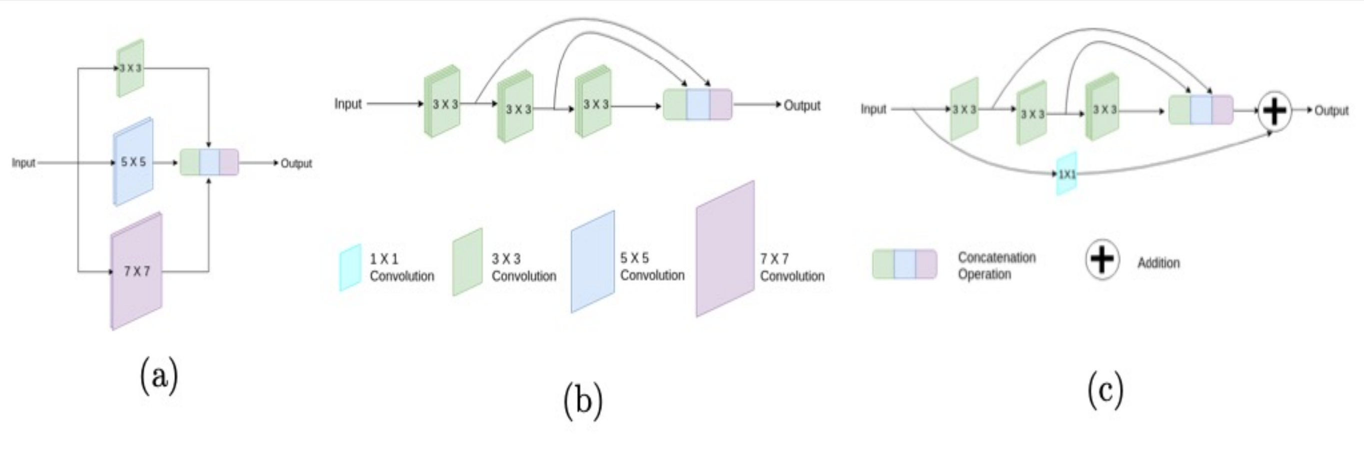
1. Among multi-scale encoder features
2. Between the stages of the encoder and decoder

➡ Find an effective way to fuse features

Introduction Unet, Unet++, MultiRes Unet



- Unet++



- MultiResUnet

- Unet++

  -> The fusion of only same-scale feature maps

- MultiResUnet

  -> Somewhat balance the possible semantic gaps

➡ Capturing multi-scale features is essential for resolving complex scale variations

Introduction Unet, Unet++, MultiRes Unet

- How to sufficiently **bridge the semantic gap between the encoder and decoder** through **multi-scale channel-wise information** fusion by effectively capturing the non-local semantic dependencies

- We rethink the skip connection design and **propose an alternative method** for better connecting the features between the encoder and decoder stages

Module 1. Channel-wise Cross Fusion Transformer (CCT)

Module 2. Channel-wise Cross Attention (CCA)

- Contributions

1. Our study is the first work that sufficiently **explores the potential weakness of skip connections** in U-Net on multiple datasets and finds that the independent simple copying is not appropriate
2. We suggest a **new perspective** to boost semantic segmentation performance
3. Our method is a more appropriate combination of U-Net and Transformer with **less computational cost and higher performance**

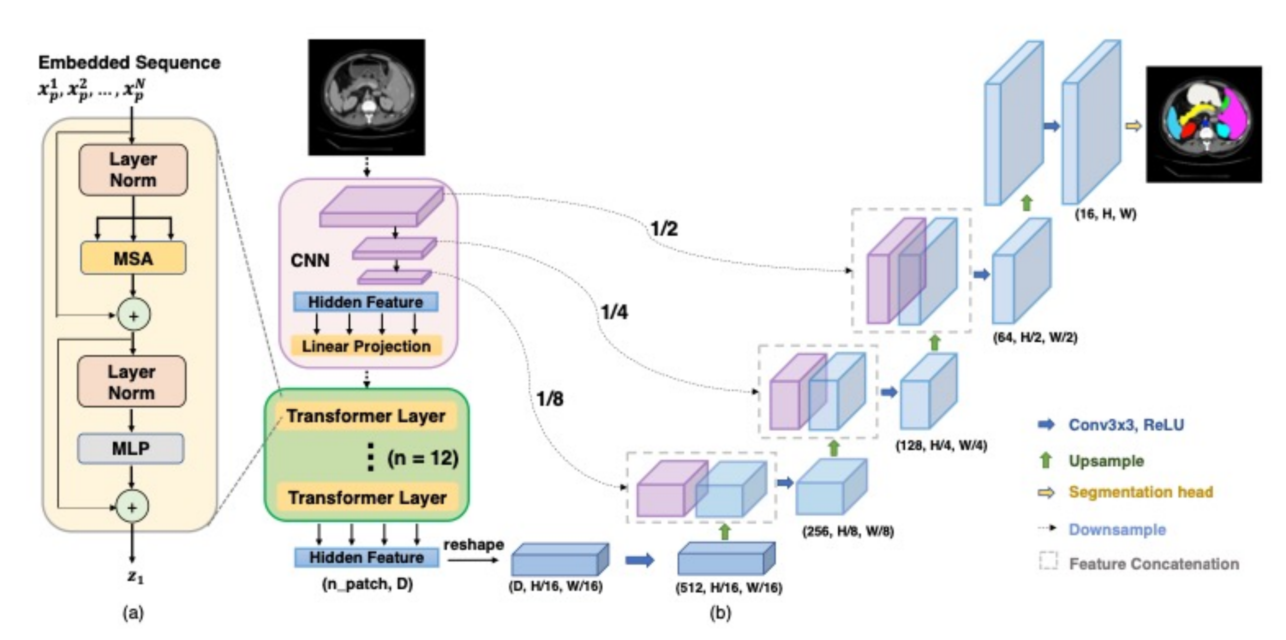
Related Work

01 Transformer based Models

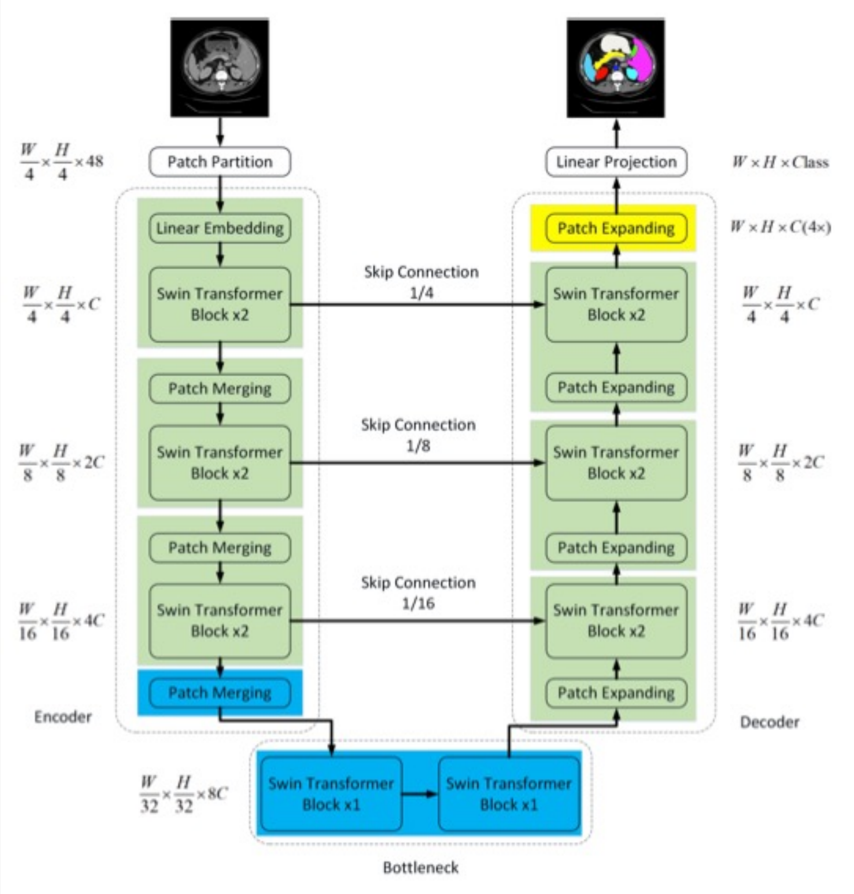
02 Skip Connections

Related Work Transformers

- TransUNet

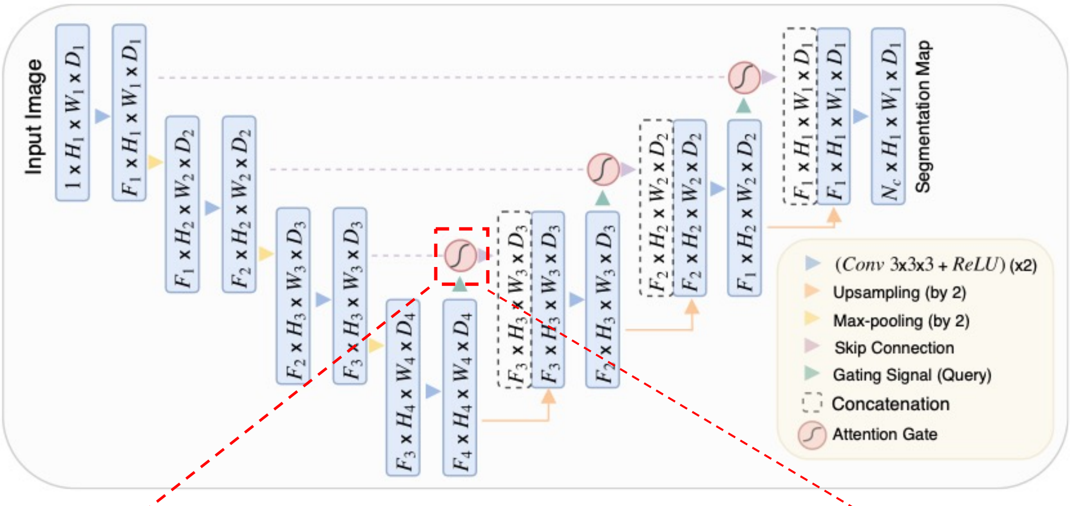


- Swin-Unet

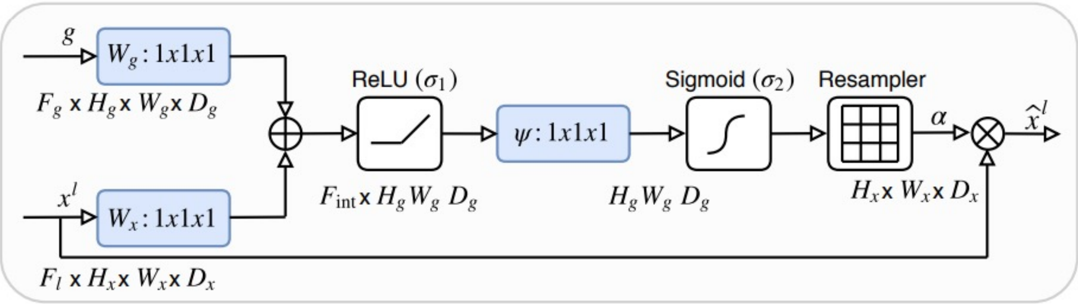


- Transformer based model의 아쉬운 점
  - ➡ Mainly focus on the defects of convolution rather than U-Net if-self

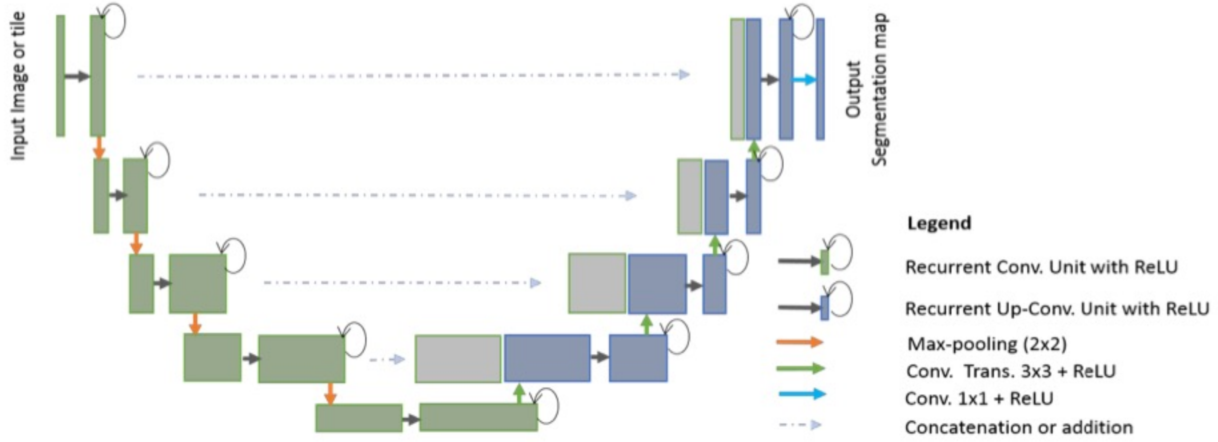
Related Work Skip Connections



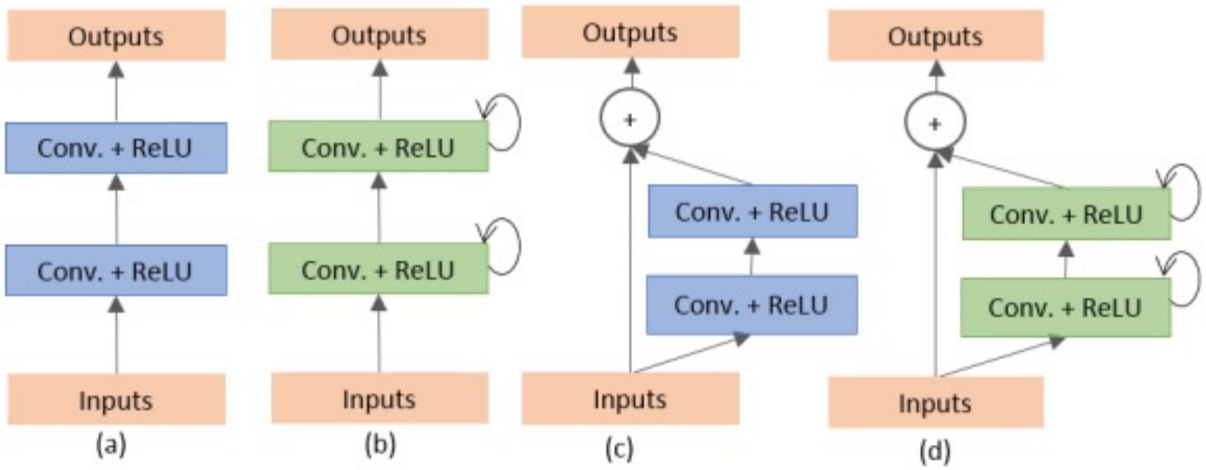
- Attention U-Net



- Attention Gate

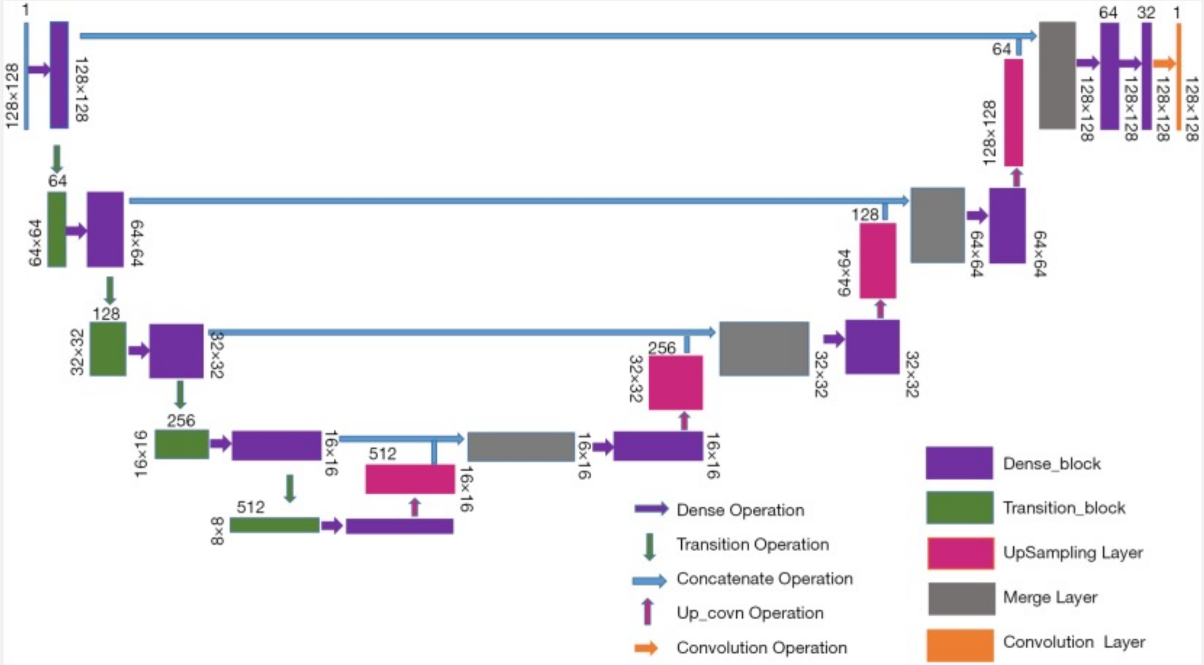


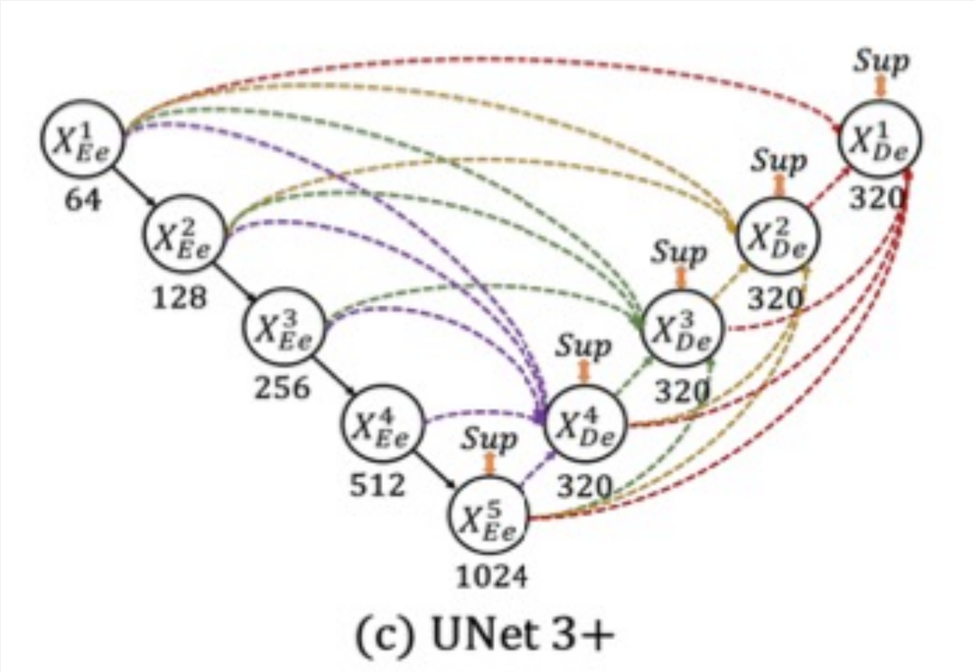
- R2U-Net



- RCL, RRCNN

Related Work Skip Connections





- DenseUNet

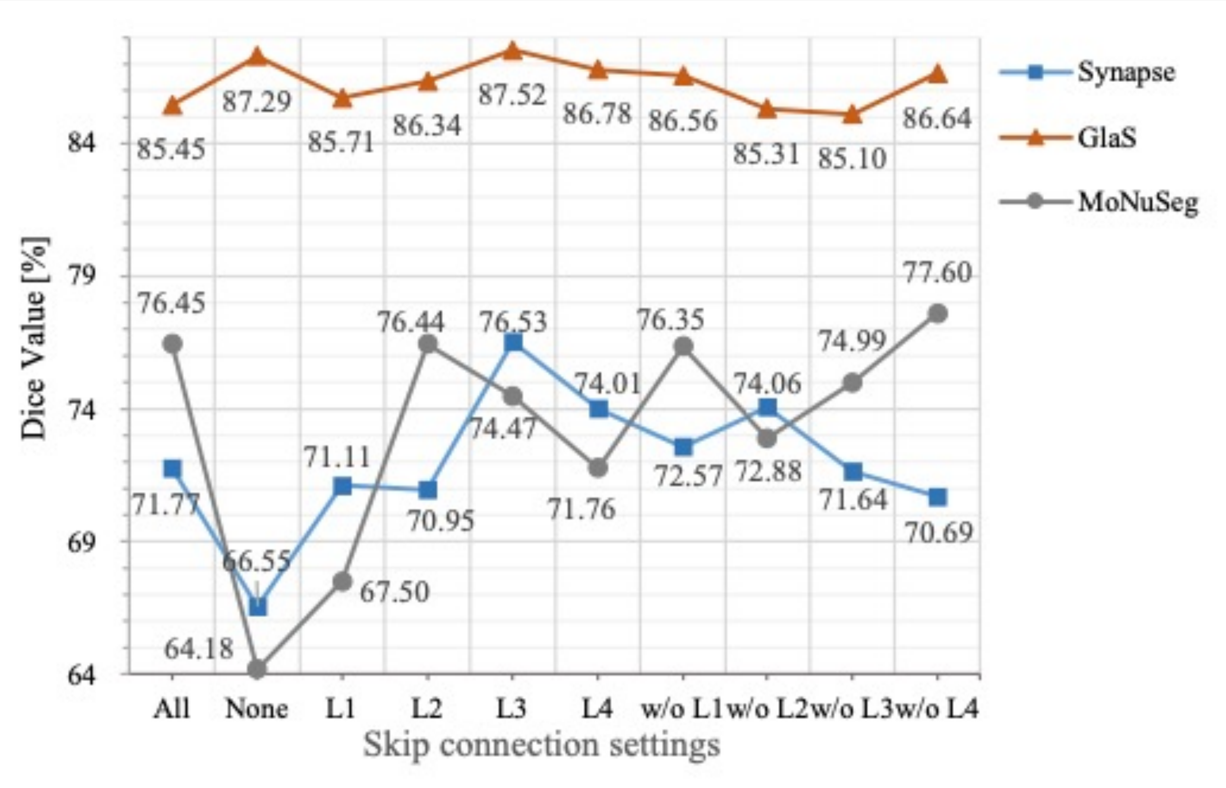
- UNet3+

- Skip connections 강화 모델의 아쉬운 점

➡ These methods assume that each skip connection has equal contribution

The Analysis of Skip connections

The Analysis of Skip connections



- Finding 1

  The U-net without any skip connection is even better than the original U-Net

- Finding 2

  Not all skip connections with simple copying are useful for segmentation

- Finding 3

  The optimal combination of skip contributions is different for different datasets

Q & A

UCTransNet for Medical Image Segmentation

01 **CCT : Channel-wise Cross Fusion Transformer for Encoder Feature Transformation**

02 **CCA: Channel-wise Cross Attention for Feature Fusion in Decoder**

UCTransNet Modules : CCT

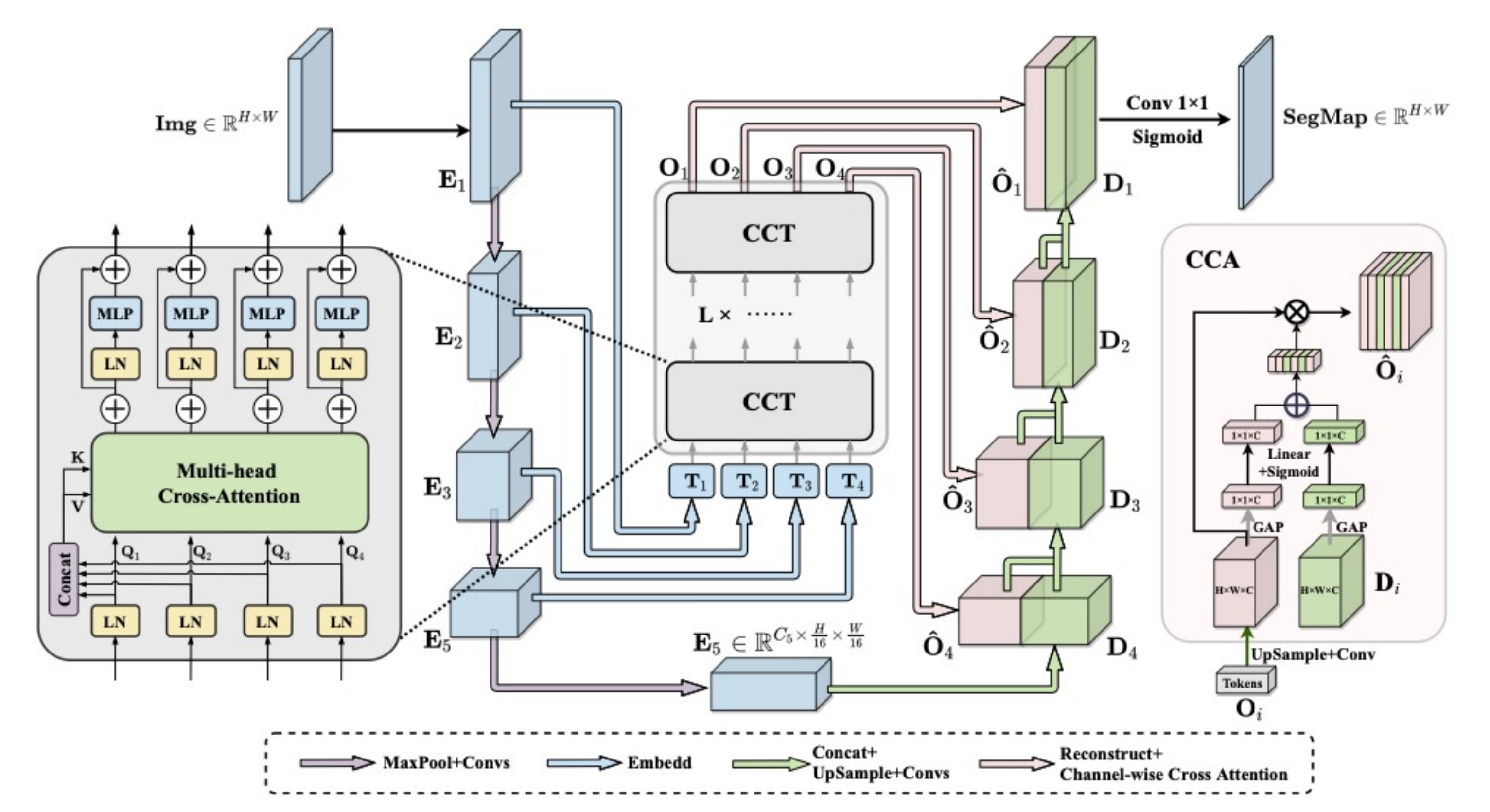
- Multi-scale Feature Embedding

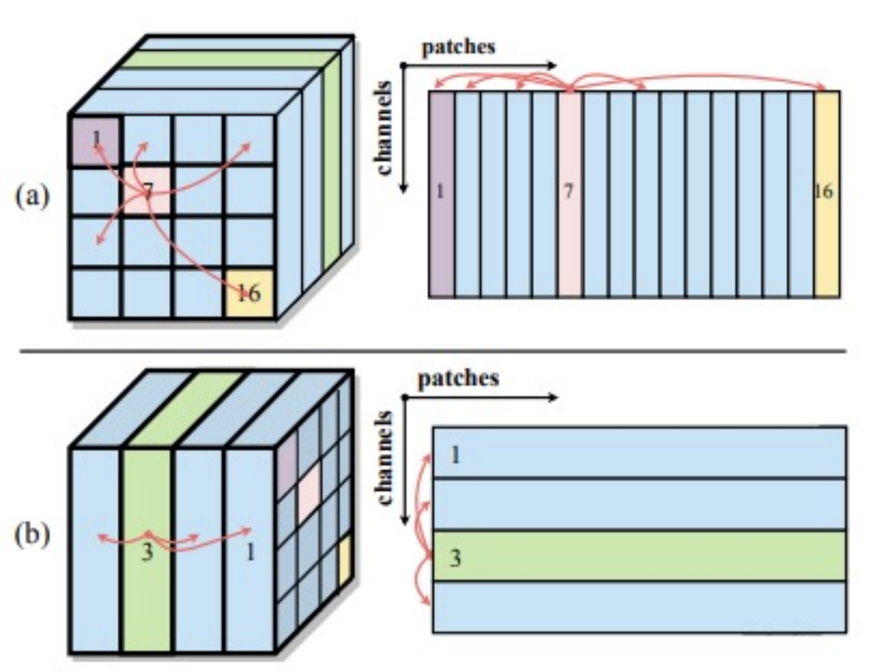
  - Outputs of Skip connection layers

  - Tokenization

$$\mathbf{E}_i \in \mathbb{R}^{\frac{HW}{i^2} \times C_i}, (i = 1, 2, 3, 4)$$

$$\mathbf{T}_i \in \mathbb{R}^{\frac{HW}{i^2} \times C_i}$$





- channel-wise cross-attention

UCTransNet Modules : CCT

- Multi-head Cross-Attention

  Query, Key, Value

$$\mathbf{Q}_i = \mathbf{T}_i W_{\mathbf{Q}_i}, \mathbf{K} = \mathbf{T}_\Sigma W_{\mathbf{K}}, \mathbf{V} = \mathbf{T}_\Sigma W_{\mathbf{V}}$$

- Cross Attention mechanism(CA)

$$\begin{aligned}\mathrm{CA}_i = \mathbf{M}_i \mathbf{V}^\top &= \sigma\left[\psi\left(\frac{\mathbf{Q}_i^\top \mathbf{K}}{\sqrt{C_\Sigma}}\right)\right]\mathbf{V}^\top \\ &= \sigma\left[\psi\left(\frac{W_{\mathbf{Q}_i}^\top \mathbf{T}_i^\top \mathbf{T}_\Sigma W_{\mathbf{K}}}{\sqrt{C_\Sigma}}\right)\right] W_{\mathbf{V}}^\top \mathbf{T}_\Sigma^\top\end{aligned}$$

$$\mathrm{MCA}_i = (\mathrm{CA}_i^1 + \mathrm{CA}_i^2 +, \ldots, +\mathrm{CA}_i^N)/N$$

$$\mathbf{O}_i = \mathrm{MCA}_i + \mathrm{MLP}(\mathbf{Q}_i + \mathrm{MCA}_i)$$

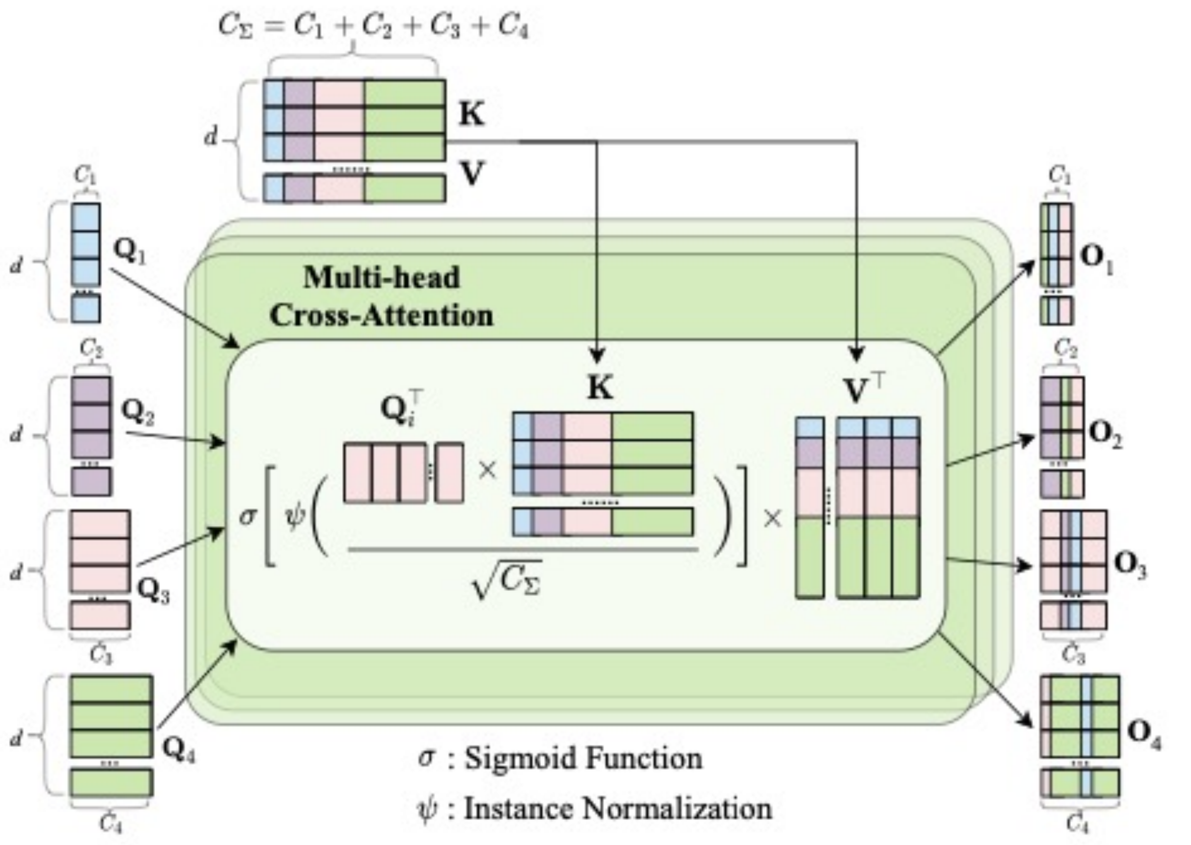


Figure 5: Multi-head Cross-Attention.

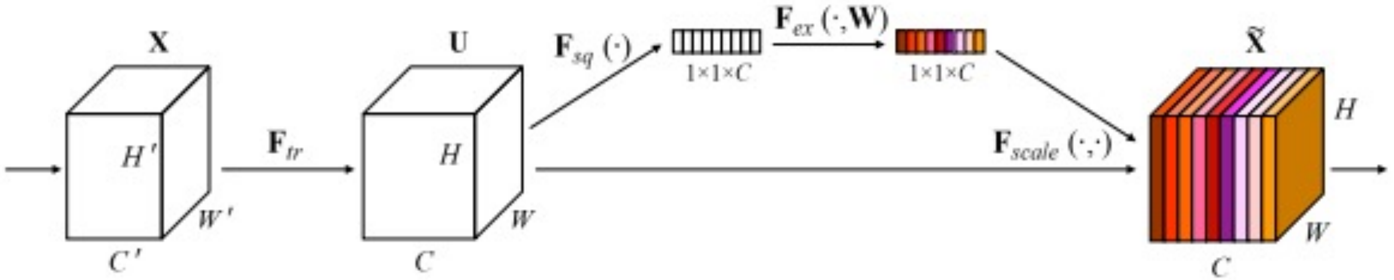
UCTransNet Modules : CCA

- Global Average Pooling

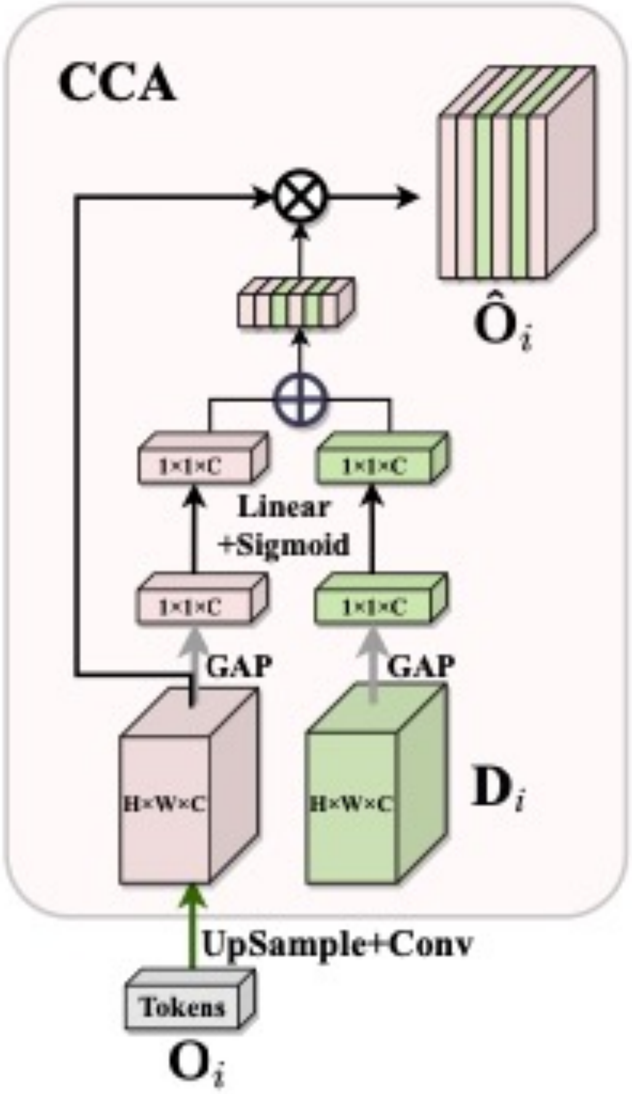
$$\mathcal{G}(\mathbf{X}) = \frac{1}{H\times W}\sum_{i=1}^{H}\sum_{j=1}^{W}\mathbf{X}^k(i,j)$$

$$\mathbf{M}_i = \mathbf{L}_1\cdot\mathcal{G}(\mathbf{O_i}) + \mathbf{L}_2\cdot\mathcal{G}(\mathbf{D_i})$$

- Squeeze-and-Excitation (참조)



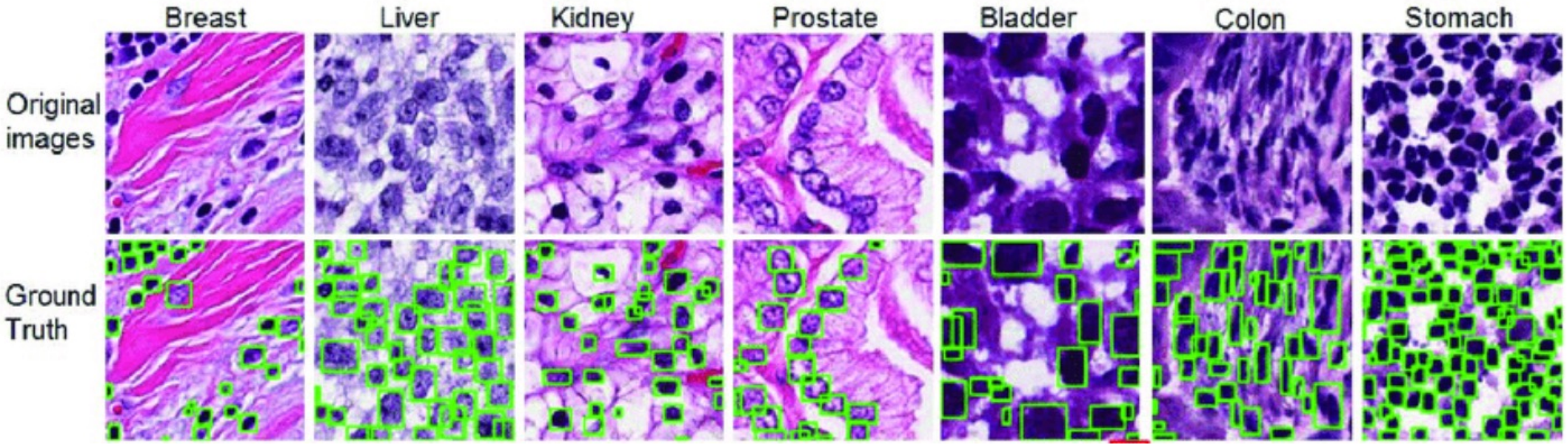
Sigmoid : to build Channel Attention Map



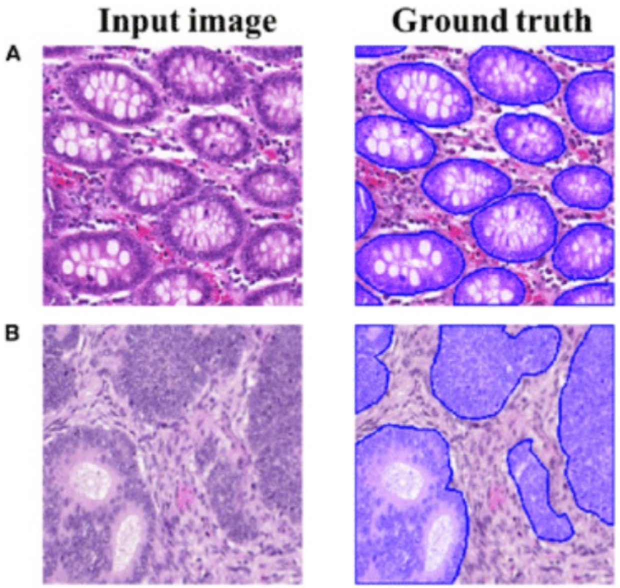
Experiments

Experiments

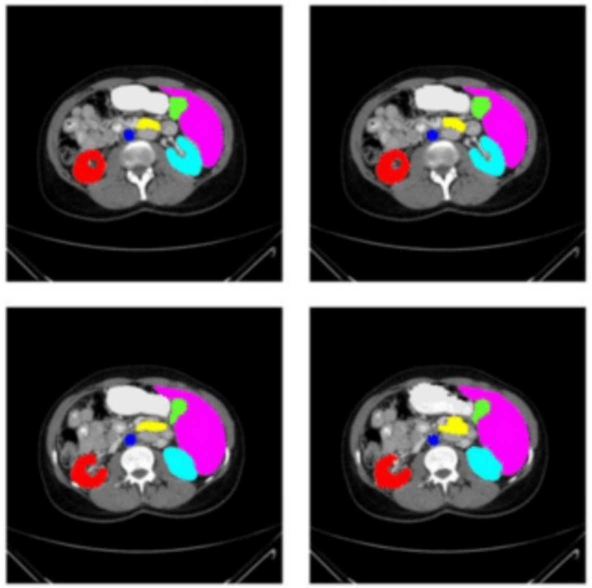
- Datasets



- MoNuSeg
  - 1000 x 1000 이미지
  - train 30, test 14
  - 7 organs
  - 세포핵 segmentation



- GlaS
  - 위장 조직검사 데이터
  - 165 데이터
  - pixel resolution of 0.465µm.



- Synapse
  - 50 데이터
  - 512x512x85
    512x512x198
  - 6 organs

Experiments

- Quantitative Inspection

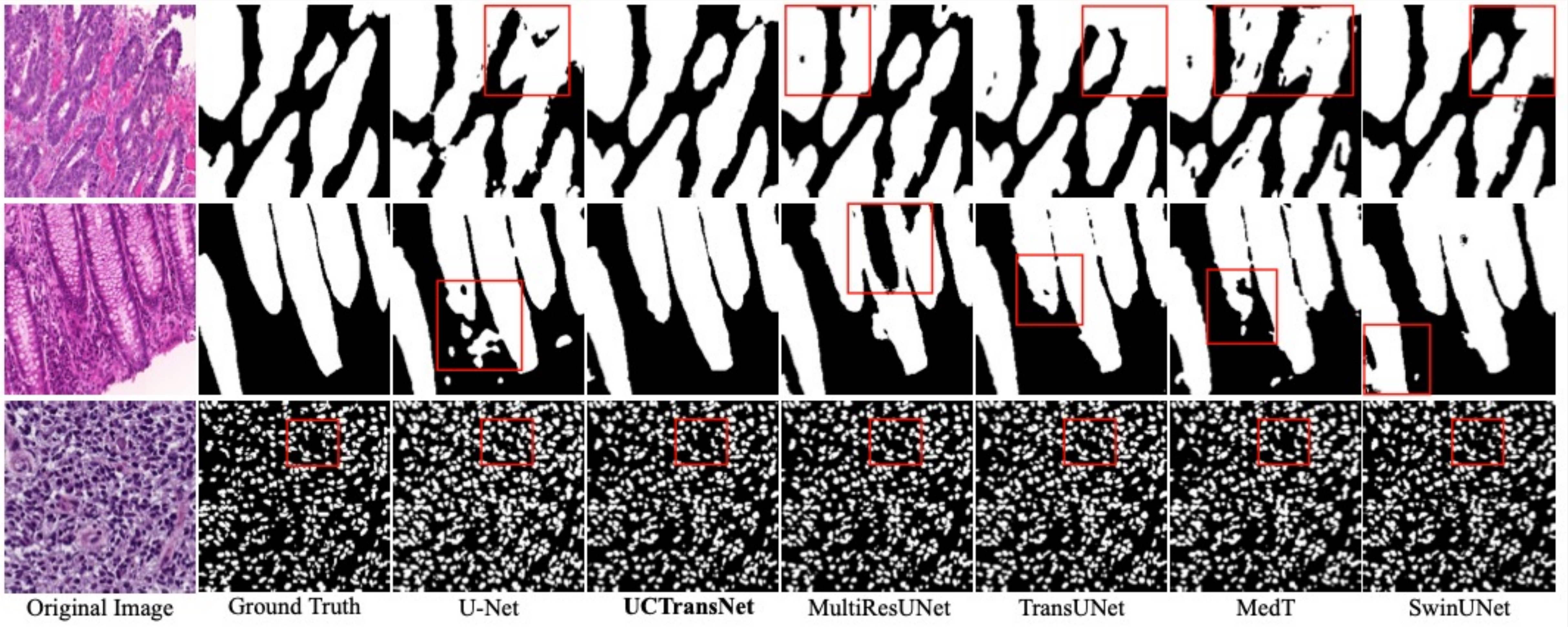
Method	Param (M)	FlOPs (G)	GlaS Dice (%)	GlaS IoU (%)	MoNuSeg Dice (%)	MoNuSeg IoU (%)
U-Net	14.8	50.3	85.45±1.25	74.78±1.67	76.45±2.62	62.86±3.00
UNet++	74.5	94.6	87.56±1.17	79.13±1.70	77.01±2.10	63.04±2.54
AttUNet	34.9	101.9	88.80±1.07	80.69±1.66	76.67±1.06	63.47±1.16
MRUNet	57.2	78.4	88.73±1.17	80.89±1.67	78.22±2.47	64.83±2.87
TransUNet	105	56.7	88.40±0.74	80.40±1.04	78.53±1.06	65.05±1.28
MedT	98.3	131.5	85.92±2.93	75.47±3.46	77.46±2.38	63.37±3.11
Swin-Unet	82.3	67.3	89.58±0.57	82.06±0.73	77.69±0.94	63.77±1.15
Ours	65.6	63.2	**90.18±0.71***	**82.96±1.06***	**79.08±0.67**	**65.50±0.91**

Methods	Dice↑	HD↓
V-Net (2016)	68.81	-
DARR (2020)	69.77	-
U-Net (2015)	71.77	53.04
R50-U-Net	74.68	36.87
R50-AttUNet (2018)	75.57	36.97
TransUNet (2021)	77.48	31.69
Swin-Unet (2021)	**79.13**	**21.55**
UCTransNet (w/o CCA)	78.99	30.29
UCTransNet-pre	75.54	38.97
UCTransNet	78.23	26.75

Table 2: Comparison with state-of-the-art segmentation methods on Synapse dataset. For simplicity, 'R50-U-Net' and 'R50-AttUNet' denote U-Net and Attention U-Net with ResNet-50 as backbone, respectively.

Experiments

- Qualitative Inspection

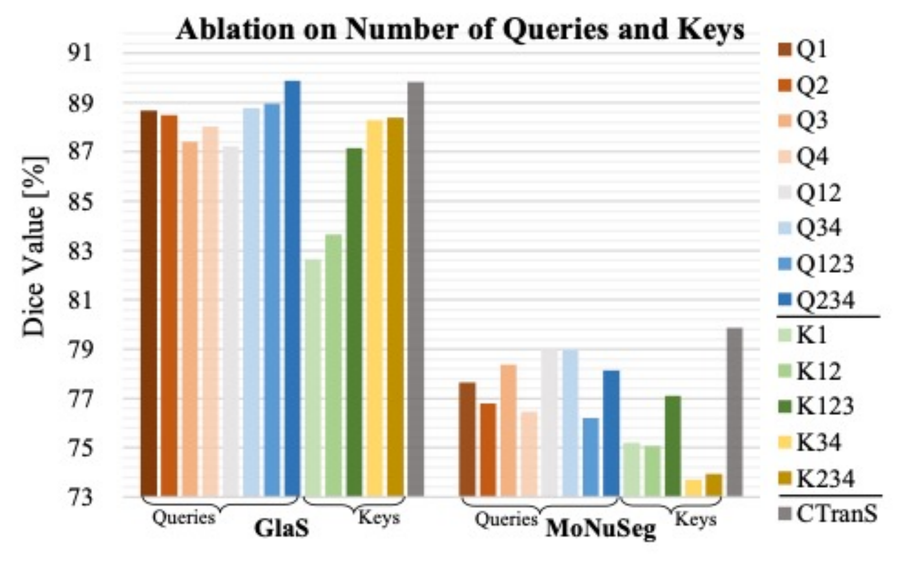


Experiments

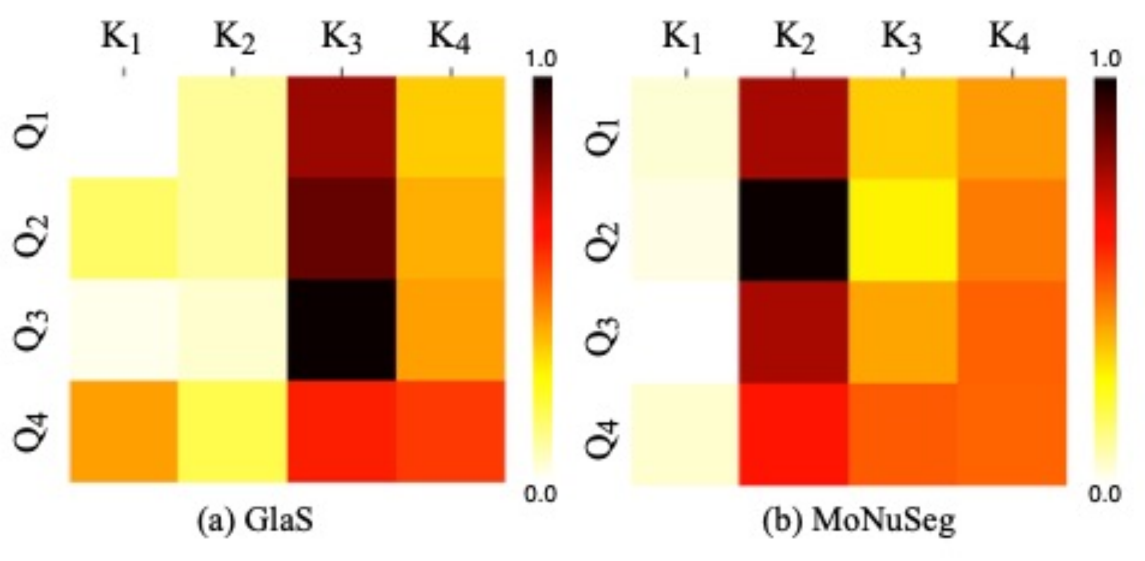
- CCT, CCA

Method	GlaS		MoNuSeg	
	Dice(%)	IoU(%)	Dice(%)	IoU(%)
Baseline (U-Net)	85.45	74.78	76.45	62.86
Baseline+CCT	89.09	80.78	79.31	65.97
Baseline+CCA	87.09	78.10	76.84	63.85
Baseline+CCT+CCA	**89.84**	**82.24**	**79.87**	**66.68**

- Ablation Studies on the Number of Queries and Keys



- The Cross Attention Matrix in CCT Module



Conclusion

1. We introduced a Channel Transformer Segmentation network (UCTransNet) from the channel-wise perspective to provide precise and reliable automatic segmentation of medical images

2. The proposed approach significantly improves the state-of-the-art results in medical image segmentation on multiple benchmark datasets.

3. The UCTransNet model indeed successfully narrows the semantic gap and takes full advantage of the multiscale features in the encoding stage

Q & A